Wave-rib sandwich pickleball paddles

PB-EL-001 Elongated (390 × 200 × 16 mm) and PB-HY-001 Hybrid (390 × 190 × 16 mm). Linear static and modal finite-element analysis.

1 Summary

Both paddles were analysed as a 4 mm carbon / 8 mm printed wave-rib / 4 mm carbon sandwich. Each was held at the grip and loaded by a 500 N ball impact, first at the sweet spot (y = 265 mm from the butt) and then at the tip (y = 345 mm). The tip impact is the worst case for every quantity.

- **The carbon face sheets and the printed core have large margins.** The carbon reaches at most 141 MPa (reserve factor ≥ 3.6). The core reaches 17.6 MPa von Mises (RF ≥ 2.6). The highest face-sheet stress is in the throat, where the 30 mm handle widens into the blade.
- **The adhesive bond between the core and the carbon is the critical item.** Bond shear peaks at 10.1 MPa on the centre lands, 15–30 mm above the solid part of the core. Across that band the 99th-percentile value is 5.5 MPa. Against the 8 MPa allowable assumed for an epoxy bond to printed PA-CF, the band passes (RF 1.45) but the local peaks do not (RF 0.79). §5 and §7 cover the cause and the fixes.
- **Stiffness.** Under 500 N the blade deflects 3.8 mm for a tip impact and 2.2 mm at the sweet spot. With the grip held, the first bending mode is 105 Hz (Elongated) and 112 Hz (Hybrid).
- **Weight.** The Elongated paddle weighs 832 g and the Hybrid 785 g, about 3.5 times a typical 210–240 g paddle. The two 4 mm carbon sheets alone weigh 608 g / 569 g. §6 has the breakdown.

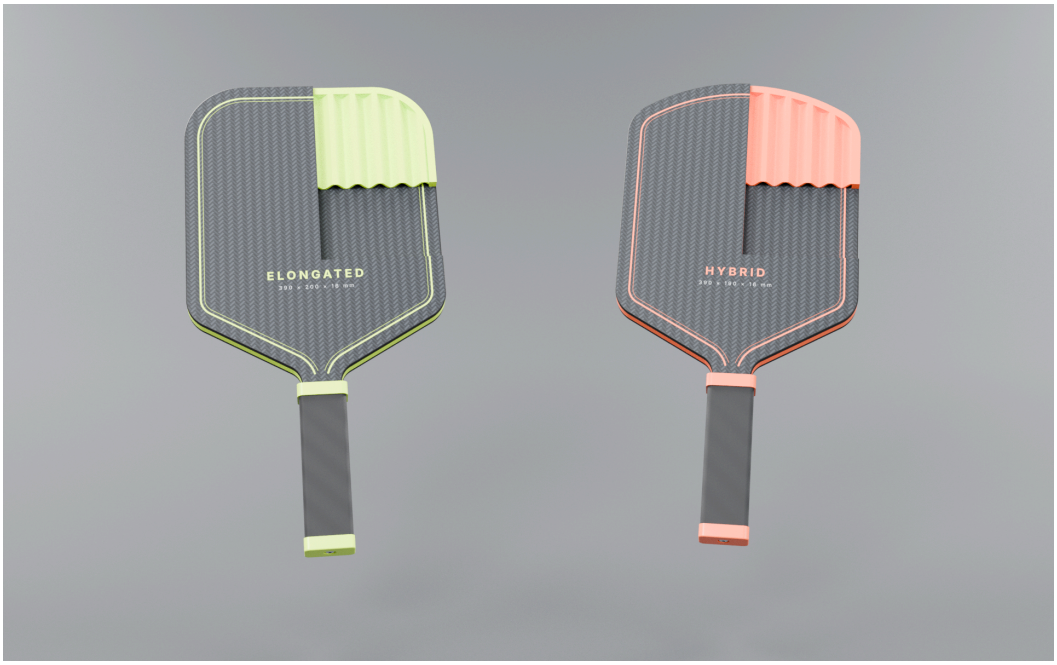
Worst case (tip impact, 500 N)	Elongated	Hybrid	Status (both)
Deflection, sweet spot / tip impact	2.25 / 3.85 mm	2.24 / 3.84 mm	—
Carbon face sheets, max fibre stress	141 MPa	137 MPa	✔ RF 3.57
Printed core, von Mises	17.5 MPa	17.6 MPa	✔ RF 2.56
Printed core, shear across the print layers	10.1 MPa	10.1 MPa	✔ RF 2.48
Bond line, peel (tension)	4.8 MPa	4.9 MPa	▲ RF 1.03
Bond line, shear: 99th percentile in the hot band	5.4 MPa	5.5 MPa	▲ RF 1.45
Bond line, shear: local peak	10.1 MPa	10.1 MPa	✖ RF 0.79
First bending mode (grip held)	105 Hz	112 Hz	—
Mass	832 g	785 g	▲ 3.5 × typical

Reserve factor RF = allowable ÷ calculated stress; RF ≥ 1 passes. ✔ means RF ≥ 1.5, ▲ means 1.0–1.5 and ✖ means below 1. The carbon RF uses the max-stress failure index (RF = 1 / index).

2 Construction analysed

- **Face sheets.** Woven carbon/epoxy plates, 4 mm thick. Top and bottom are identical and cover the full outline, blade and handle.
- **Core.** 8 mm high, 3D-printed in PA-CF (carbon-filled nylon), printed flat. A closed 6 mm perimeter frame covers the whole edge. Inside it is a sinusoidal wave rib: pitch 18 mm, wall 1.6 mm, amplitude 4 mm. The flutes run along the paddle. The crests are cut flat where they meet the sheets, which gives bond lands 4.25 mm wide.
- **Solid zone.** The core is solid from the butt to y = 150 mm (handle and throat). Each flute starts with a 45° run-out, 8 mm long, which prints without support. The butt end holds an M4 heat-set insert for the butt-cap screw.

- **Grip.** Sleeve, wrap, throat collar and butt cap are not load-carrying in this model. The hand is represented by a clamp (§3).



Cutaway render: the top sheet and the core are cut back to show the wave rib between the sheets.

3 Finite-element model

- **Geometry.** Half model, symmetric about the centre line. Top sheet, core and bottom sheet are meshed together, so the bond is perfect and shares nodes. The core is modelled exactly as drawn: frame, solid zone, run-outs, wave walls and lands.
- **Mesh.** 10-node quadratic tetrahedra (C3D10). Element size is 1.6 mm in the core, which puts at least two elements through every wave wall, and 3.5 mm in the sheets. Meshed in gmsh, solved in CalculiX 2.21.
- **Support.** The outer surfaces of the handle from y = 14 to 112 mm are fixed, which stands in for the hand on the grip. Next to a rigid clamp the stresses are artificially high, so results closer than 8 mm to its edge (y < 120 mm) are not assessed. Table 2 lists them for reference.
- **Load.** 500 N spread uniformly over a Ø30 mm patch (the ball contact) on the top sheet, at LC1 (x = 0, y = 265) or LC2 (x = 0, y = 345). The load is applied as a static peak force. A 26 g ball that changes speed by about 33 m/s during a 4 ms contact gives about 215 N on average and about 340 N peak. The 500 N used is about 1.5 times that, to cover harder hits and some dynamic amplification.
- **Modal.** Natural frequencies with the same grip support, on a coarser 3 / 5 mm mesh.

Model	Nodes	Elements	Degrees of freedom	Load patch area (half)
ELONGATED	305,534	175,706	916,602	342 mm²
HYBRID	288,217	165,800	864,651	350 mm²

Materials and allowables

Material	Stiffness	Allowables (design values)
Woven carbon/epoxy plate 0/90 fabric, ρ = 1.55 g/cm³	$E_1 = E_2 = 60$ GPa, $E_3 = 8$ GPa $G_{12} = 4.5$ GPa, $G_{13} = G_{23} = 3.5$ GPa $\nu_{12} = 0.05$, $\nu_{13} = \nu_{23} = 0.3$	Tension 600 MPa, compression 500 MPa In-plane shear 90 MPa Interlaminar shear 55 MPa
Printed PA-CF core solid walls, ρ = 1.18 g/cm³	E = 4 GPa, ν = 0.35 (isotropic)	von Mises 45 MPa Across the print layers 25 MPa
Epoxy bond, carbon to printed core	— (perfect bond)	Shear 8 MPa, peel 5 MPa Conservative values for abraded and primed nylon. Confirm them by coupon tests (§7).

These are typical design values, not supplier data. When the actual carbon plate, filament and adhesive are chosen, replace them with the supplier's values. Every reserve factor scales directly with its allowable.

4 Results

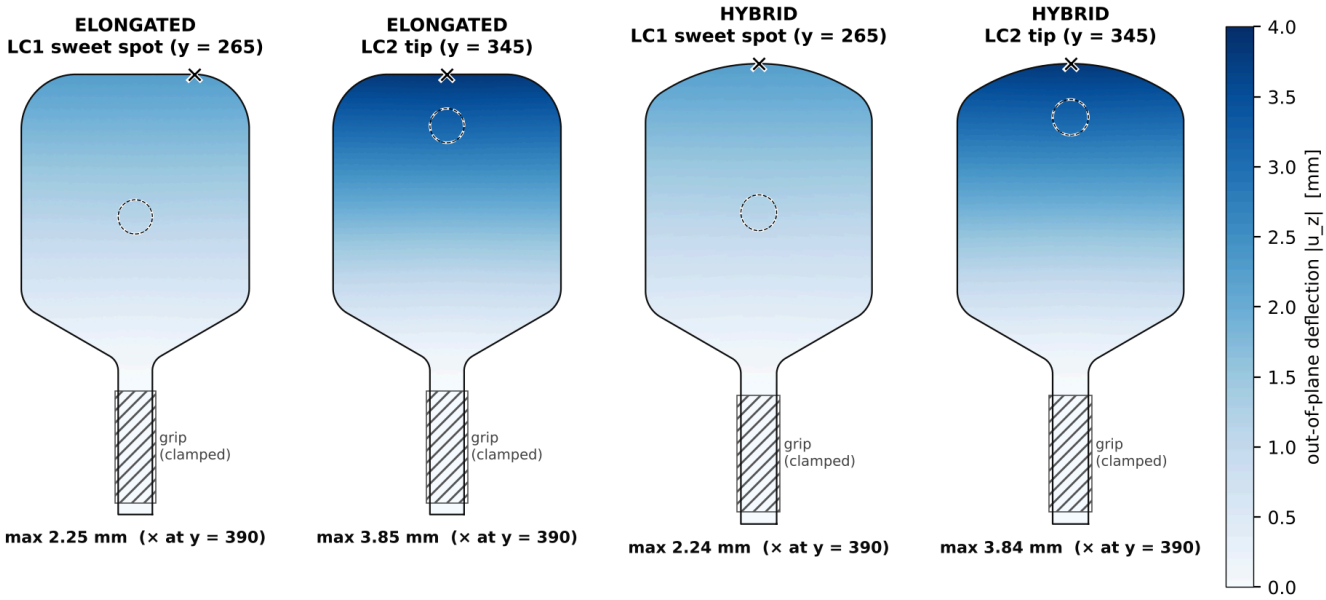


Fig. 1 – Out-of-plane deflection under 500 N. Dashed circle: ball contact patch. Hatched: grip clamp.

Table 2 – all results	Allowable	Elongated LC1	Elongated LC2	Hybrid LC1	Hybrid LC2
Max deflection [mm]	—	2.25	3.85	2.24	3.84
Carbon: fibre stress |σ11|, |σ22| [MPa]	600 t / 500 c	88	141	85	137
Carbon: in-plane shear τ12 [MPa]	90	12.8	20.9	12.9	20.9
Carbon: interlaminar shear [MPa]	55	3.5	7.0	3.4	6.8
Carbon: max-stress failure index	1.00	0.18	0.28	0.17	0.27
· carbon fibre stress at the clamp edge (not assessed)		125	186	125	186
Core: von Mises [MPa]	45	12.0	17.5	12.3	17.6
Core: through-thickness compression under the ball [MPa]	45	6.1	6.3	4.6	5.6
Core: shear across the print layers $\sqrt{(\tau xz^2+\tau yz^2)}$ [MPa]	25	6.9	10.1	7.1	10.1
Bond: shear, local peak [MPa]	8	6.9	10.1	7.1	10.1
· at height y [mm]		165	165	176	176
Bond: shear, 99th percentile in the 10 mm band at the peak [MPa]	8	3.7	5.4	3.9	5.5
Bond: shear, mean over the whole bond [MPa]		0.38	0.70	0.40	0.73
Bond: peel, local peak [MPa]	5	3.4	4.8	3.4	4.9
Bond: peel, 99th percentile in the band at the peak [MPa]	5	2.5	3.5	2.5	3.5

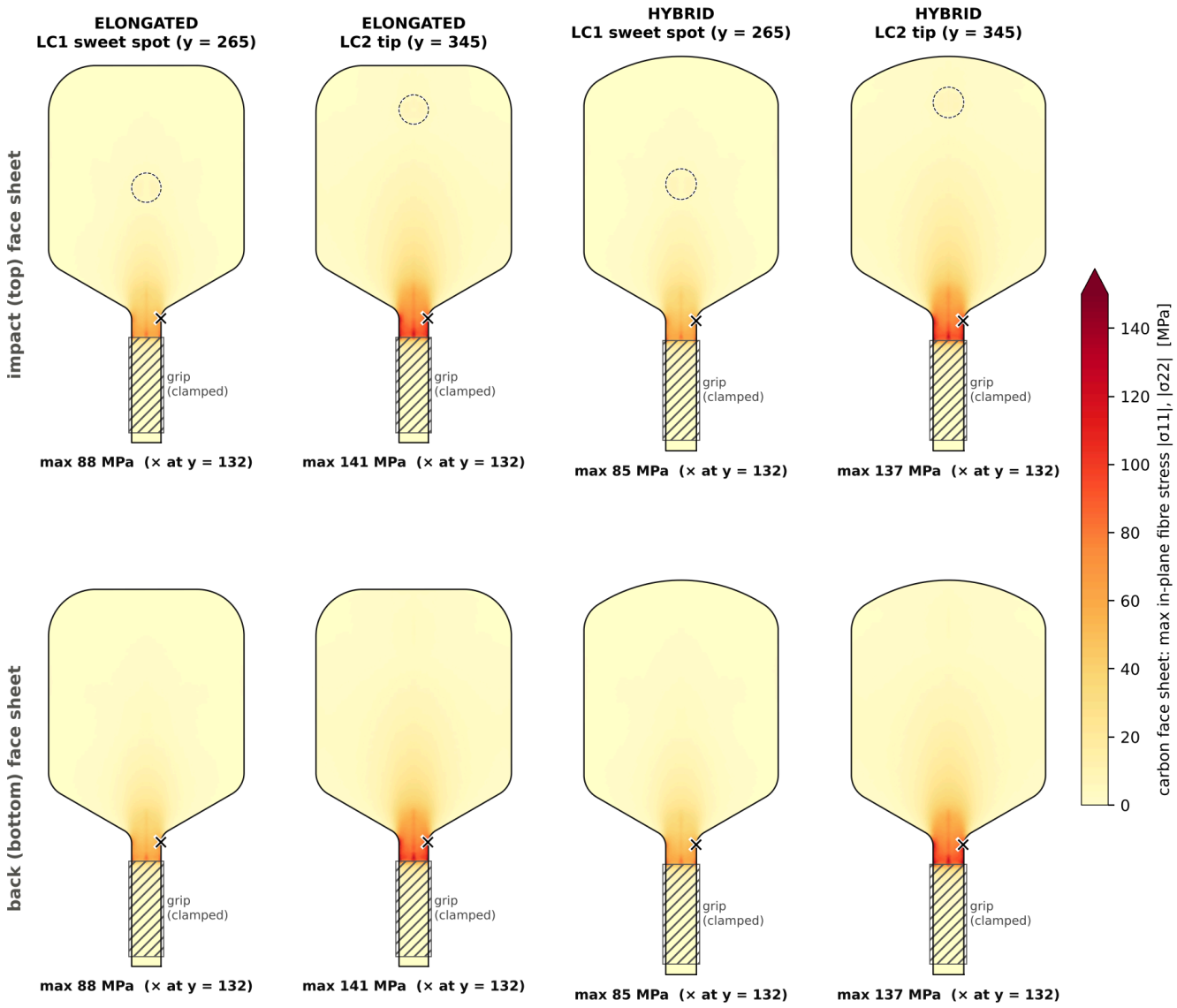


Fig. 2 – Carbon face sheets, largest in-plane fibre stress. The stress concentrates in the throat; the colour scale saturates at the clamp edge, which is not assessed.

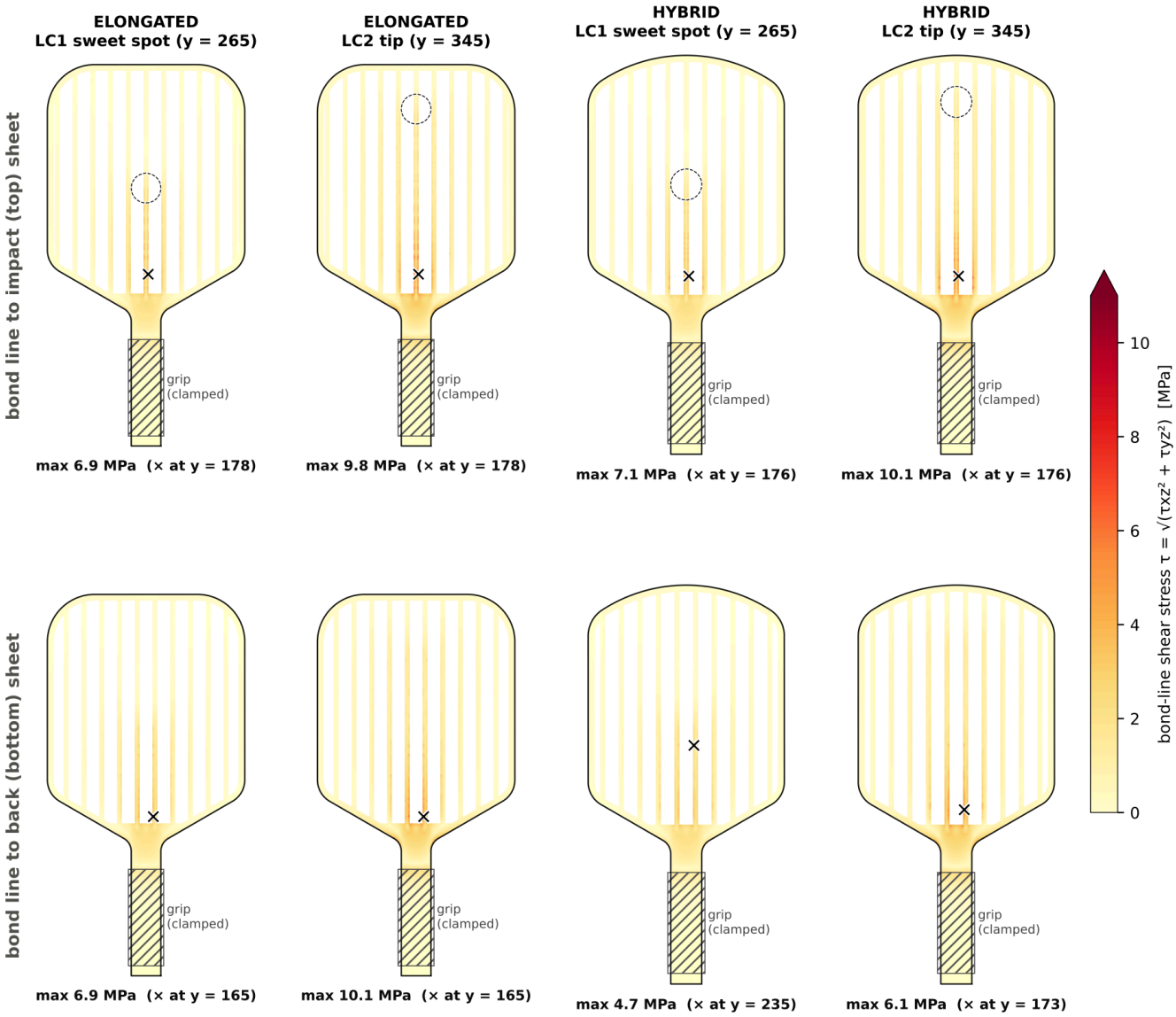


Fig. 3 – Shear stress in the bond line, plotted on the core's bonded surfaces: the 6 mm frame, the solid zone and the 4.25 mm lands. White stripes are the open flutes.

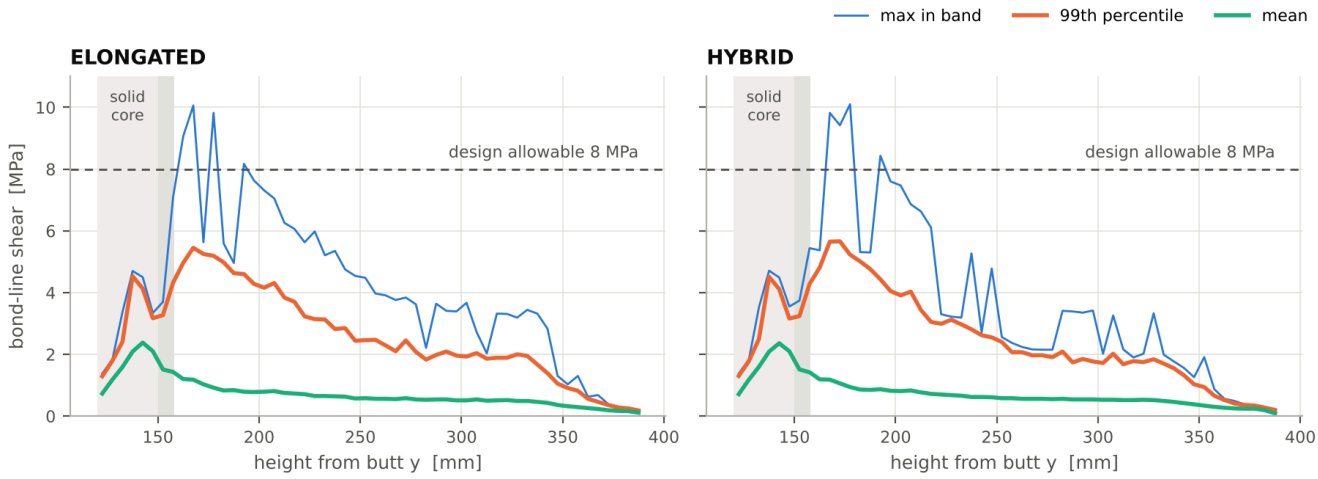


Fig. 4 – Bond shear along the paddle for the tip impact, in 5 mm bands: largest value, 99th percentile and mean. Shaded: solid core and the 8 mm run-out.

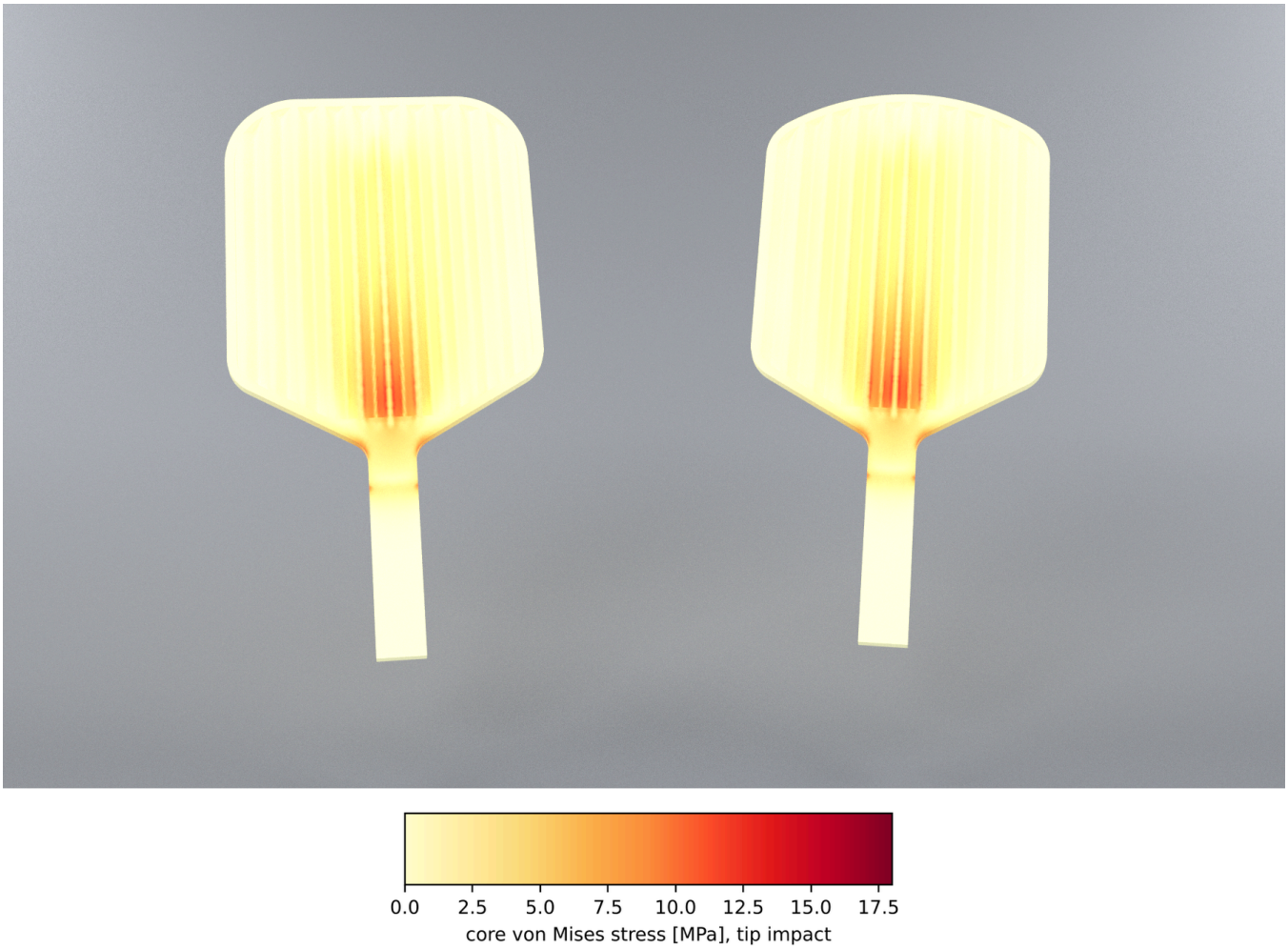


Fig. 5 – Printed core coloured by von Mises stress for the tip impact, top sheet removed. Left: Elongated, right: Hybrid.

5 Design iteration: flute ends and land width

The first version of the core had square flute ends at y = 150 mm and lands 3.13 mm wide (amplitude 3.6 mm). Its bond peeled at the step where each flute starts: 15.2 MPa against a 5 MPa allowable. The final core starts each flute with a 45° run-out, so the core meets the sheet in a thin wedge instead of a step. Its amplitude is 4 mm, which widens the lands to 4.25 mm. Elongated, tip impact:

	First version	Final
Bond peel, local peak	15.2 MPa at y = 150	4.8 MPa at y = 157
Bond peel, 99th percentile in the band at the peak	8.1 MPa	3.5 MPa
Bond shear, local peak	6.6 MPa at y = 150	10.1 MPa at y = 165
Bond shear, 99th percentile in the band at the peak	4.8 MPa	5.4 MPa
Bond reserve factor, local peaks (shear and peel)	✖ RF 0.33	✖ RF 0.80
Bond reserve factor, 99th percentile in the band	✖ RF 0.62	▲ RF 1.43
Core von Mises	17.8 MPa	17.5 MPa
Carbon fibre stress	143 MPa	141 MPa
Tip deflection	3.91 mm	3.85 mm

The peel peak has gone. The shear peaks now sit on the centre lands just above the solid zone. There the shear from the whole blade funnels into the 30 mm wide throat. With the wider lands, the wall under each land

is thinner (0.8 mm instead of 1.2 mm), which sharpens those local peaks. Across that band the 99th-percentile level rose a little, from 4.8 to 5.4 MPa. Fig. 4 shows the peaks are narrow spikes and that the 99th-percentile line stays below the allowable along the whole blade. Spikes like these depend on the mesh and on the sharp corner where the wall meets the sheet. In the real part, the adhesive fillet at the land edge rounds that corner off.

6 Mass, balance and swing weight

Part (from the CAD model)	Elongated	Hybrid
Carbon face sheet, top (4 mm)	304.2 g	284.6 g
Wave-rib core with frame (PA-CF)	178.6 g	170.2 g
Carbon face sheet, bottom (4 mm)	304.2 g	284.6 g
Grip sleeve (TPU)	26.3 g	26.3 g
Grip wrap, PU (0.75 mm)	9.5 g	9.5 g
Throat collar (TPU)	1.7 g	1.7 g
Butt cap (TPU)	5.0 g	5.0 g
Screw ISO 7380 M4 × 12	1.8 g	1.8 g
Heat-set insert M4 (brass)	1.1 g	1.1 g
Total	**832 g**	**785 g**
Balance point from the butt	239 mm	234 mm
Swing weight about a pivot 100 mm from the butt	241 kg·cm²	216 kg·cm²

A typical competition paddle weighs 210–240 g (7.5–8.5 oz). Its swing weight is roughly 100–130 kg·cm² about the same pivot. These paddles are about 3.5 times as heavy and about twice the swing weight. The 4 mm carbon sheets make up 73 % of the mass. The printed core weighs 179 g, an effective density of about 455 kg/m³. A polypropylene honeycomb core is typically 80–120 kg/m³. The margins in §4 show how much strength there is to give up.

7 Recommendations

Bond line (before the first build)

- Use a toughened structural epoxy or MMA adhesive that is rated for nylons. Abrade both surfaces, then clean and prime them as the adhesive supplier specifies. Leave a small fillet of adhesive along every land edge.
- Test single-lap shear coupons of carbon plate bonded to printed PA-CF (ISO 4587 / ASTM D1002 layout) and flatwise-tension coupons (ASTM C297). Use the measured strengths as the bond allowables in this analysis.
- If the coupons give less than about 12 MPa in shear, lower the peak at the start of the wave. Either thicken the wall under the centre lands (wall 2.0 mm instead of 1.6 mm), or extend the solid zone up the centre of the blade by about 30 mm, then re-run the analysis.

Weight

- For a playable paddle, the sheets and the core both need to be much lighter. One option is 0.5–1.0 mm carbon sheets on a 14–15 mm core of honeycomb, foam or a much thinner-walled print, which keeps the 16 mm thickness. Because a sandwich gets its bending stiffness from the distance between the sheets, thin sheets on a thicker core can stay stiff. All the geometry is parametric (paddle/geometry.py), so a lighter lay-up can be re-analysed with the same tools.

- If the 4/8/4 build is meant as a prototype or a training paddle, the structure as analysed is strong enough. The bond line is the only item to qualify.

Validation on the first parts

- Static stiffness check: clamp the grip and hang a known load at the tip. The analysis predicts 0.77 mm per 100 N at y = 345 mm for the Elongated and 0.77 mm per 100 N for the Hybrid.
- Tap test: the first bending mode should ring at about 105 Hz (Elongated) and 112 Hz (Hybrid), with the handle held firmly.
- Check for bond-line voids (tap or ultrasound) and for cracks at the throat after impact tests.

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8 Assumptions and limitations

- Linear elastic, small deflection, quasi-static peak load. No fatigue, no damage growth, no strain-rate effects.
- The bond is perfect. The adhesive layer is not modelled; bond stresses are read in the core elements next to each sheet.
- The printed core is treated as isotropic and fully dense. Real prints are weaker and softer across the layers (covered by the separate 25 MPa allowable) and depend on the print settings.
- The grip is a rigid clamp. A real hand and grip are softer, which lowers the stress at the throat slightly and raises the deflection slightly. Edge guard, paint and grip parts are not modelled.
- Peak values at sharp re-entrant corners depend on the mesh. Percentile and band values (Fig. 4) are the robust measures.